

SynthDrive: An Actuarially Structured Synthetic Telematics Generator for Motor Insurance Research

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Abstract

Access to real telematics data for academic and independent insurance research is severely restricted by proprietary and contractual barriers, and the only widely cited public benchmark—the So-Boucher-Valdez (2021) dataset—was generated by an opaque SMOTE-plus-neural-network pipeline that is hard to calibrate or extend. SynthDrive v0.1 is an open-source Python package that addresses this gap with a transparent three-stage pipeline: a Gaussian copula for correlated feature generation, a zero-inflated negative binomial model with an exposure offset for claim frequency, and a Gamma model for aggregate severity. All stages enforce actuarial and physical constraints, including compositional day-of-week proportions, monotone hard-event threshold sequences, and claim-amount consistency. Validated against the So-Boucher-Valdez (SBV) seed dataset at 100,000 policies, SynthDrive reproduces marginal distributions and generalized linear model (GLM) coefficient directions within acceptable tolerance. The package requires no GPU hardware, ships with a built-in validation report and a formal algebraic specification, and is provided with full data provenance disclosure: SynthDrive is calibrated from a public synthetic seed, not from real insurer data.

JEL Codes: G22, C46, C88, C15, R41

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1 Introduction

Usage-based insurance (UBI) is a motor insurance pricing methodology that links premium to observed driving behavior through telematics devices, mobile applications, or on-board diagnostic ports. The resulting data—mileage, hard-braking events, acceleration counts, time-of-day exposure, and turn intensity—offer substantially richer risk signals than traditional rating variables such as driver age, vehicle age, and territory. Several major insurers in North America and Europe have deployed UBI at scale since at least the early 2000s, and the academic literature on telematics-based pricing has grown correspondingly (Bolderdijk et al., 2011; Paefgen et al., 2014; Lemaire et al., 2016; Verbelen et al., 2018; Boucher et al., 2020).

Despite this growth, independent quantitative research on UBI remains constrained by a persistent data access problem. Real telematics portfolios are proprietary. Regulators and insurers are

reluctant to share policy-level telematics records, and the few datasets that have appeared in the literature were released under restricted agreements that prevent redistribution. The practical result is that researchers outside of insurer partnerships cannot reproduce published results, cannot benchmark new methods against a common dataset, and cannot develop and test pricing or fraud models on realistic telematics structure.

The only widely cited remedy to date is the public synthetic dataset of [So et al. \(2021\)](#), hereafter the SBV dataset, which contains 100,000 synthetic motor insurance policies with traditional rating variables, a rich telematics feature set, claim counts, and aggregate claim amounts. It was generated by a three-stage process: SMOTE-based portfolio resampling, feedforward neural network classification for claim counts, and feedforward neural network regression for claim amounts. The SBV dataset is a valuable public resource, but its generation pipeline is opaque and difficult to inspect or re-calibrate; no open validation framework was released alongside it; and the method cannot be easily adapted to produce portfolios with different sizes, market calibrations, or schema variants without retraining the underlying network.

SynthDrive v0.1 addresses these limitations with a simple design principle: replace the neural network generator with a transparent, classical statistical pipeline whose every assumption is documented, testable, and modifiable. The core components are a Gaussian copula for joint feature generation, a zero-inflated negative binomial model for claim frequency, and a Gamma model for aggregate severity. The pipeline enforces the actuarial and physical constraints that any credible telematics dataset must satisfy, and produces a reproducible portfolio from a single integer seed.

The package is released as open-source Python. This paper serves as its methodological companion. Section 2 provides background on UBI data, the SBV dataset, and related synthetic tabular data methods. Section 3 describes the SynthDrive v0.1 architecture in full. Section 4 documents data provenance. Section 5 presents the validation results. Section 6 states the limitations of the current work. Section 7 concludes.

2 Background and Related Work

This section reviews UBI data structure, the SBV benchmark dataset, and the main families of synthetic tabular data generation methods, before identifying the actuarial constraints that distinguish telematics data from generic tabular datasets.

2.1 Usage-based insurance and telematics data structure

A personal auto telematics dataset at the policy level typically contains three groups of variables. *Traditional rating variables*—driver age, sex, marital status, vehicle age and use, territory, and years without claims—underpin standard actuarial GLM rate-making and are well-understood in the pricing literature ([de Jong and Heller, 2008](#)). *Telematics variables* are generated by the monitoring device or application and commonly include annual mileage driven, the percentage of annual mileage recorded, the average number of days per week driven, proportions of driving by day of week and time of day, hard-acceleration and hard-braking event counts at several threshold speeds (in mph/s), and lateral turn-intensity event counts at several g-force thresholds; [Boucher et al. \(2020\)](#) survey the actuarial evidence on which of these variables carry incremental predictive power beyond the traditional factors. *Claim variables* form the outcome layer: claim count $N_i \in \{0, 1, 2, \dots\}$ and

aggregate claim amount $A_i \geq 0$, with the constraint $A_i = 0$ whenever $N_i = 0$. Pure premium is defined as $PP_i = A_i/e_i$, where e_i is the policy’s exposure in policy-years. The frequency-severity decomposition is standard in actuarial modeling (Klugman et al., 2012) and ensures that zero-inflation in claim counts and the heavy-tailed distribution of non-zero severities are handled by separate, appropriate distributional families.

2.2 The So–Boucher–Valdez dataset

So et al. (2021) generated a 100,000-policy synthetic telematics dataset to serve as a public benchmark for UBI research, drawing on a proprietary portfolio from The Co-operators General Insurance Company, a Canadian insurer, which cannot be shared publicly. The generation process has three stages: a synthetic feature portfolio is produced by an extended SMOTE algorithm applied to the real feature space; claim counts are generated as successive binary classifications using feedforward neural networks; and aggregate claim amounts are generated by a feedforward neural network regression with claim count included as a feature. The resulting 52-column dataset has been used in several published studies on telematics pricing and synthetic data benchmarking. The authors validated it by fitting Poisson and Gamma regression models to both the real and synthetic portfolios and comparing estimated coefficients, reporting directional consistency across the main rating factors.

For the purposes of SynthDrive, the SBV dataset serves as the calibration seed: the most credible public telematics benchmark available, and the closest achievable proxy to real insurer data without a data-sharing agreement.

2.3 Synthetic tabular data: a brief taxonomy

The literature on synthetic tabular data generation has grown substantially since the introduction of CTGAN (Xu et al., 2019). The main methodological families are copula-based, GAN-based, and diffusion-based. Copula-based methods model the joint distribution of continuous variables by separating dependence structure from marginal distributions (Nelsen, 2006); Gaussian copulas are the simplest and most interpretable member of this family and have a long history in actuarial applications (Frees and Valdez, 1998). GAN-based methods such as CTGAN (Xu et al., 2019) and CTAB-GAN (Zhao et al., 2021) apply conditional generative adversarial networks to mixed-type tabular data and can model complex non-linear dependence, but are training-intensive and can be unstable. Diffusion-based methods, including TabDDPM (Kotelnikov et al., 2023) and TabSyn (Zhang et al., 2024), are currently state-of-the-art on general tabular generation benchmarks but require GPU training and offer less interpretability than copula-based approaches.

SynthDrive uses a Gaussian copula because it is fully interpretable, computationally light, and well-suited to the actuarial constraint structure of telematics data, where marginal distributions are known from the seed and the principal modeling object is the rank correlation structure. TabSyn and TabDDPM are reserved for SynthDrive v0.2 as an advanced generation option, not a replacement for the core pipeline.

2.4 Actuarial constraints in synthetic telematics data

Generic tabular generators do not enforce the constraint structure that actuarial telematics data requires. The five constraints that SynthDrive enforces are:

Compositional constraint. Daily driving proportions (Monday through Sunday) must sum to unity; weekend and weekday aggregate proportions are derived quantities, not free variables.

Monotone hard-event sequences. Hard-braking and hard-acceleration counts must be non-increasing as the speed threshold increases, because a higher-threshold event is a subset of lower-threshold events.

Claim consistency. Aggregate claim amount must be zero whenever claim count is zero, and positive otherwise.

Exposure offset. Claim frequency must be modeled with an exposure offset so that policies of different durations are compared on a common annualized basis.

Physical bounds. Driver age, vehicle age, mileage, credit score, and all proportion variables must lie within physically meaningful ranges.

Enforcing these constraints is straightforward in a classical parametric pipeline but requires custom post-processing layers in deep generative models.

3 SynthDrive v0.1: Architecture and Methods

This section describes the SynthDrive v0.1 generation pipeline in full, covering feature generation, claim frequency, claim severity, constraint enforcement, and the complete output schema.

3.1 Overview

SynthDrive v0.1 generates a synthetic telematics portfolio of n policies through a sequential three-stage pipeline. Each policy record contains continuous and categorical driver and vehicle features, an exposure measure, derived telematics variables, a claim count, an aggregate claim amount, and a pure premium. The three stages are:

1. **Feature generation:** a Gaussian copula produces the continuous feature vector X_i^c ; categorical variables X_i^d are sampled from empirical proportions; exposure e_i is derived from a duration draw; and derived telematics variables T_i are constructed from X_i^c and e_i .
2. **Claim frequency:** a zero-inflated negative binomial (ZINB) model produces claim count N_i conditional on X_i^c , X_i^d , and e_i .
3. **Claim severity:** a Gamma model produces aggregate claim amount A_i conditional on $N_i > 0$, X_i^c , and X_i^d .

A deterministic constraint-enforcement pass (Section 3.7) is applied after stage 1 and again after stage 3. The entire pipeline is seeded by a single integer `random_state` passed to NumPy’s `default_rng`. In full, SynthDrive v0.1 produces a portfolio $\{(X_i^c, X_i^d, e_i, T_i, N_i, A_i, PP_i)\}_{i=1}^n$.

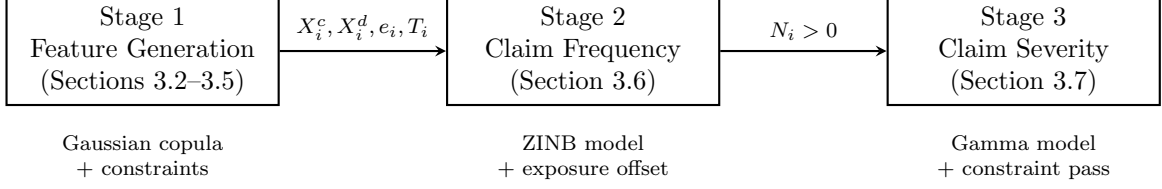


Figure 1: SynthDrive v0.1 three-stage generation pipeline. Arrows show the outputs passed between stages; constraint enforcement is applied after stages 1 and 3.

The complete pipeline is exposed through a two-function public API:

```

from synthdrive import generate, validate

df = generate(n=100_000, preset="core", random_state=42)
report = validate(df)

```

3.2 Continuous feature generation: Gaussian copula

The nine continuous variables that drive both the feature set and the claim model are assembled into a copula vector for policy i :

$$X_i^c = (\text{age}_i, \text{credit}_i, \text{yncl}_i, \text{cage}_i, \text{miles}_i, \text{days}_i, \text{pct}_i, \text{accel}_{9,i}, \text{brake}_{9,i})^\top.$$

The acceleration and braking variables are anchor counts at the 9 mph/s threshold; the full six-threshold sequences are derived from these anchors in Section 3.4. Car age is defined on $[-2, 20]$ to match the SBV data, where negative values reflect vehicles registered before the policy start year. Accel and brake anchors are raw event counts, not rates per mile.

The joint dependence structure is modeled by a Gaussian copula. Let $\Sigma^S \in [-1, 1]^{9 \times 9}$ be the target Spearman rank correlation matrix, estimated from the SBV data in seed mode or taken from calibrated defaults in parameter mode. The corresponding Pearson correlation matrix is obtained element-wise by:

$$\rho_{jk}^P = 2 \sin\left(\frac{\pi \rho_{jk}^S}{6}\right), \quad (1)$$

which is the exact relationship between Spearman and Pearson correlations for a bivariate normal pair (Kruskal, 1958). If the resulting matrix Σ^P is not positive definite, the nearest positive-definite correlation matrix is found via eigenvalue flooring at $\epsilon = 10^{-8}$ followed by diagonal rescaling (Higham, 2002).

Sampling proceeds via the Cholesky factorization $\Sigma^P = LL^\top$:

$$Z_i \sim \mathcal{N}(0, I_9), \quad (2)$$

$$\tilde{Z}_i = LZ_i, \quad (3)$$

$$U_{ij} = \Phi(\tilde{Z}_{ij}), \quad j = 1, \dots, 9, \quad (4)$$

where Φ is the standard normal CDF. The vector U_i has uniform marginals on $(0, 1)$ and rank correlations that approximate Σ^S . Each continuous variable is then recovered as $X_{ij}^c = F_j^{-1}(U_{ij})$, where F_j is the marginal CDF for variable j . In seed mode, F_j^{-1} is the empirical quantile function estimated from the SBV data; in parameter mode, the parametric marginals in Table 1 are used.

Table 1: Parametric marginal distributions for the nine continuous copula variables (parameter mode).

Variable	Distribution	Key parameters	Support
Driver age (age)	Truncated normal	$\mu = 45, \sigma = 14$	$[16, 90]$
Credit score (credit)	Truncated normal	$\mu = 650, \sigma = 85$	$[400, 900]$
Years no claims (yncl)	Exponential	mean = 4	$[0, 50]$
Car age (cage)	Shifted gamma	shape = 2.5, scale = 3.2, shift = -2	$[-2, 20]$
Annual miles (miles)	Log-normal	$\mu_{\log} = 9.35, \sigma_{\log} = 0.42$	$[500, 60,000]$
Avg. days/week (days)	Truncated normal	$\mu = 4.6, \sigma = 1.2$	$[0, 7]$
Annual pct. driven (pct)	Beta	$\alpha = 2.2, \beta = 4.5$	$(0, 1)$
Accel anchor (accel ₉)	Log-normal (rounded)	$\mu_{\log} = 3.18, \sigma_{\log} = 1.20$	$\mathbb{Z}_{\geq 0}$
Brake anchor (brake ₉)	Log-normal (rounded)	$\mu_{\log} = 1.79, \sigma_{\log} = 0.85$	$\mathbb{Z}_{\geq 0}$

3.3 Categorical variables and territory

The discrete feature vector $X_i^d = (\text{sex}_i, \text{marital}_i, \text{use}_i, \text{terr}_i, \text{reg}_i)$ is sampled independently of X_i^c , which is a simplifying assumption reflecting the weak rank correlations observed between categorical and continuous variables in the SBV data. All values are stored as strings matching the SBV column values exactly. Empirical proportions from the SBV dataset ($n = 100,000$) are used in both modes:

$$\begin{aligned}
\text{sex}_i &\sim \text{Categorical}(p_{\text{Male}} = 0.539, p_{\text{Female}} = 0.461), \\
\text{marital}_i &\sim \text{Categorical}(p_{\text{Married}} = 0.699, p_{\text{Single}} = 0.301), \\
\text{use}_i &\sim \text{Categorical}(p_{\text{Commute}} = 0.498, p_{\text{Private}} = 0.461, p_{\text{Commercial}} = 0.026, p_{\text{Farmer}} = 0.014), \\
\text{reg}_i &\sim \text{Categorical}(p_{\text{Urban}} = 0.781, p_{\text{Rural}} = 0.219).
\end{aligned}$$

Territory terr_i is a continuous integer zone code with observed range $[11, 91]$, mean 56.5, and standard deviation 24.0 in the SBV data; it is not a categorical variable. In both modes, it is sampled from the empirical discrete distribution of zone codes estimated from the SBV data.

3.4 Duration, exposure, and derived telematics variables

Duration and exposure. In the SBV dataset, duration is recorded as an integer number of days. Exposure in policy-years is derived as $e_i = \text{Duration}_i / 365$. In parameter mode, duration is drawn from the discrete mixture:

$$\text{Duration}_i \sim 0.60 \cdot \delta_{365} + 0.30 \cdot \delta_{182} + 0.10 \cdot \delta_{91},$$

where δ_v denotes a point mass at v days.

Total miles driven. This is a derived variable combining annual mileage, exposure, and the fraction of mileage actually recorded:

$$\text{total_miles}_i = \text{miles}_i \cdot e_i \cdot (0.6 + 0.8 \cdot \text{pct}_i) \cdot \exp(\varepsilon_i^m), \quad \varepsilon_i^m \sim \mathcal{N}(0, 0.12^2).$$

Day-of-week proportions. The seven daily proportions $(p_{\text{mon},i}, \dots, p_{\text{sun},i})$ are sampled from a Dirichlet distribution with concentration vector α^{dow} calibrated from the SBV data, so that the compositional constraint $\sum_d p_{d,i} = 1$ is satisfied by construction. Weekday and weekend aggregates are derived quantities:

$$p_{\text{wkday},i} = p_{\text{mon},i} + p_{\text{tue},i} + p_{\text{wed},i} + p_{\text{thr},i} + p_{\text{fri},i}, \quad p_{\text{wkend},i} = p_{\text{sat},i} + p_{\text{sun},i}.$$

Thursday is stored as `Pct.drive.thr` following SBV column naming.

Time-of-day proportions. Five time-of-day proportion variables are sampled independently from Beta distributions calibrated to the SBV data: $p_{\text{rush_am}}, p_{\text{rush_pm}}, p_{\text{2hrs}}, p_{\text{3hrs}}, p_{\text{4hrs}}$.

Hard-event threshold sequences. Hard-braking and hard-acceleration counts are derived from the copula anchor counts. For each policy i and threshold k in $\mathcal{K} = \{6, 8, 9, 11, 12, 14\}$ mph/s, the raw acceleration count is:

$$\tilde{a}_{k,i} = \text{accel}_{9,i} \cdot r_k^{\text{acc}} \cdot \exp(\varepsilon_{k,i}), \quad \varepsilon_{k,i} \sim \mathcal{N}(0, 0.15^2), \quad (5)$$

rounded to the nearest non-negative integer, where r_k^{acc} is a pre-calibrated ratio for threshold k relative to the 9 mph/s anchor. The non-increasing physical constraint is enforced by a cumulative minimum:

$$\text{accel}_{k,i} = \text{cummin}_{k \in \mathcal{K}}(\tilde{a}_{k,i}). \quad (6)$$

The same construction applies to braking counts. There is no 10 mph/s column in the SBV data; the threshold set is exactly $\{6, 8, 9, 11, 12, 14\}$.

Turn intensity. Turn intensity columns are raw event counts, not proportions or rates. The SBV data shows mean values of approximately 916 and 2,006 for left and right turn intensity at the 0.8 g threshold respectively, with maxima in the hundreds of thousands. Base counts $\text{lt}_{08,i}$ and $\text{rt}_{08,i}$ are sampled from log-normal marginals via a bivariate Gaussian copula, then sub-threshold counts at $\{0.8, 0.9, 1.0, 1.1, 1.2\}$ g are derived analogously to (5)–(6).

Table 2: Column schema of a SynthDrive generated dataset.

Group	Columns	n
Policy and exposure	policy_id, duration, exposure	3
Driver	insured_age, insured_sex, marital_status, credit_score, years_no_claims	5
Vehicle	car_age, car_use, territory, region, annual_miles_drive	5
Telematics — mileage	annual_pct_driven, total_miles_driven, avg_days_week	3
Telematics — day of week	pct_drive_mon through pct_drive_sun, pct_drive_wkday, pct_drive_wkend	9
Telematics — time of day	pct_drive_rush_am, pct_drive_rush_pm, pct_drive_2hrs, pct_drive_3hrs, pct_drive_4hrs	5
Hard acceleration counts	accel_6_miles through accel_14_miles (thresholds 6, 8, 9, 11, 12, 14 mph/s)	6
Hard braking counts	brake_6_miles through brake_14_miles (thresholds 6, 8, 9, 11, 12, 14 mph/s)	6
Left turn intensity	left_turn_intensity_08 through left_turn_intensity_12	5
Right turn intensity	right_turn_intensity_08 through right_turn_intensity_12	5
Claim variables	claim_count, claim_amount, pure_premium	3
Total		55

3.5 Claim frequency: ZINB model

Motor insurance claim counts exhibit two behavioral regimes: low-risk policyholders who rarely if ever file a claim, and high-risk policyholders whose elevated exposure generates a heavy right tail. The zero-inflated negative binomial (ZINB) model (Minami et al., 2007) captures both — the structural-zero component addresses the former, and the overdispersed negative binomial count component addresses the latter.

Formally, the claim count N_i follows:

$$N_i = (1 - S_i) \cdot \tilde{N}_i, \quad (7)$$

where $S_i \sim \text{Bernoulli}(\pi_i)$ is the structural-zero indicator and $\tilde{N}_i \sim \text{NegBin}(\mu_i, \alpha)$ is a negative binomial variate with mean μ_i and overdispersion parameter $\alpha = 0.30$.

The zero-inflation probability is:

$$\text{logit } \pi_i = \gamma_0 + \gamma_y \cdot \text{yncl}_i + \gamma_c \cdot \frac{\text{credit}_i - 650}{100}, \quad (8)$$

with $\gamma_0 = -1.735$, $\gamma_y = 0.035$, $\gamma_c = 0.12$. The value $\text{expit}(-1.735) \approx 0.15$ gives the structural-zero probability for a reference driver with no prior claims and average credit.

The negative binomial mean uses a log link with an exposure offset:

$$\log \mu_i = \log e_i + \log \lambda_0 + f(X_i), \quad (9)$$

where $\lambda_0 = 0.0165$ is the base annual claim rate and $f(X_i)$ is the additive log-relative-risk score:

$$\begin{aligned} f(X_i) = & \beta_{\text{yng}} \cdot \mathbf{1}[\text{age}_i < 25] + \beta_{\text{sen}} \cdot \mathbf{1}[\text{age}_i > 65] + \beta_{\text{male}} \cdot \mathbf{1}[\text{sex}_i = \text{Male}] \\ & + \beta_{\text{sngl}} \cdot \mathbf{1}[\text{marital}_i = \text{Single}] + \beta_{\text{cred}} \cdot \frac{\text{credit}_i - 650}{100} \\ & + \beta_{\text{com}} \cdot \mathbf{1}[\text{use}_i = \text{Commercial}] + \beta_{\text{far}} \cdot \mathbf{1}[\text{use}_i = \text{Farmer}] \\ & + \beta_{\text{mi}} \cdot (\log \text{miles}_i - \log 12,000) + \beta_{\text{pct}} \cdot (\text{pct}_i - 0.33) + \beta_{\text{cage}} \cdot (\text{cage}_i - 9) \\ & + \beta_{\text{brk}} \cdot [\log(1 + \text{brake}_{9,i}) - \log(1 + 2)] + \beta_{\text{acc}} \cdot [\log(1 + \text{accel}_{9,i}) - \log 1] + \delta(\text{terr}_i). \end{aligned} \quad (10)$$

The $\log(1+\cdot)$ transformation is applied to the hard-event counts before centering to prevent extreme raw counts (observed up to 800+) from producing unbounded risk scores. The territory effect $\delta(\text{terr}_i)$ is a step function calibrated from the SBV data.

The calibrated coefficients are given in Table 3.

Table 3: Frequency model log-relative-risk coefficients.

Symbol	Description	Value
λ_0	Base annual claim rate	0.0165
α	Overdispersion parameter	0.30
β_{yng}	Young driver (age < 25)	-0.28
β_{sen}	Senior driver (age > 65)	+0.40
β_{male}	Male (vs. female)	+0.04
β_{sngl}	Single (vs. married)	+0.10
β_{cred}	Per 100 credit points above 650	-0.28
β_{com}	Commercial vehicle use	+0.35
β_{far}	Farmer vehicle use	-0.53
β_{mi}	Log annual miles above log 12,000	+0.22
β_{pct}	Annual pct. driven above 0.33	+4.20
β_{cage}	Vehicle age in years above 9	-0.07
β_{brk}	Log hard braking at 9 mph/s, centred	+0.25
β_{acc}	Log hard acceleration at 9 mph/s, centred	+0.18
$\delta(\cdot)$	Territory log-relative-risk (zone code)	calibrated

Because $\alpha = 0.30$ is non-integer, sampling uses the exact gamma-Poisson mixture: $\tilde{\lambda}_i \sim \text{Gamma}(1/\alpha, \alpha\mu_i)$, then $\tilde{N}_i \sim \text{Poisson}(\tilde{\lambda}_i)$.

3.6 Claim severity: Gamma model

Aggregate severity is modeled only for policies with at least one claim. For $N_i = n \geq 1$:

$$A_i \mid N_i = n, X_i \sim \text{Gamma}(\kappa \cdot n, \theta \cdot g_i), \quad (11)$$

where $\kappa = 2$ is the shape parameter per claim and $\theta = 1,567$ is the base scale parameter. The shape scales linearly with claim count, so that $\mathbb{E}[A_i \mid N_i = n, X_i] = \kappa n \theta g_i$. For the reference driver ($g_i = 1$) with a single claim, the expected severity is $\kappa \theta = 3,134$.

The risk-adjusted scale factor is:

$$\log g_i = 0.20 \cdot \mathbf{1}[\text{age}_i < 25] + 0.12 \cdot (\log \text{miles}_i - \log 12,000) + 0.018 \cdot (\text{cage}_i - 7). \quad (12)$$

A minimum claim amount of $A_i \leftarrow \max(A_i, 1)$ is applied post-sampling for all policies with $N_i \geq 1$. For $N_i = 0$, $A_i = 0$ unconditionally.

3.7 Constraint enforcement

After all sampling is complete, the following constraints are enforced deterministically in the order listed.

Scalar bounds. Exposure $e_i \in (0, 1]$; driver age $\in [16, 90]$; credit score $\in [400, 900]$; years no claims ≥ 0 ; car age $\in [-2, 20]$; annual miles $\in [500, 60,000]$; all proportion variables $\in [0, 1]$; all hard-event and turn-intensity counts $\in \mathbb{Z}_{\geq 0}$.

Compositional constraint. $\sum_{d \in \{\text{mon}, \dots, \text{sun}\}} p_{d,i} = 1$. If any row sum is zero, the vector is replaced by the uniform distribution $p_{d,i} = 1/7$.

Monotone threshold sequences. A cumulative minimum is applied from threshold 6 to 14 mph/s for acceleration and braking arrays, and from 0.8 g to 1.2 g for turn intensity, as in (6).

Claim consistency. $A_i = 0$ whenever $N_i = 0$; $A_i \geq 0$ always.

Pure premium recomputation. $\text{PP}_i = A_i/e_i$ is recomputed after all of the above.

4 Data Provenance

Precise data provenance is an intellectual honesty requirement for any synthetic data tool. This section states the provenance chain for SynthDrive v0.1 explicitly.

The SBV seed dataset is itself synthetic. [So et al. \(2021\)](#) generated it from a proprietary telematics portfolio held by The Co-operators General Insurance Company, a Canadian insurer, which cannot be shared publicly. The public SBV dataset approximates the statistical properties of the real portfolio as reported by the authors, but it is a generated artifact, not a sample from the real data.

SynthDrive’s relationship to the seed is therefore second-order: it is a synthetic generator calibrated from a synthetic seed. In seed mode, SynthDrive estimates copula parameters and marginal distributions from the public SBV dataset. In parameter mode, the calibrated defaults were derived from the same source during development. In neither mode does SynthDrive have access to real insurer

data. The correct characterization of SynthDrive output is synthetic telematics data generated by a copula–ZINB–Gamma pipeline calibrated from a publicly available synthetic telematics seed. It should not be described as representing real driving behavior, real Canadian driving behavior, or real insurer loss experience.

5 Validation

A core requirement for any synthetic data tool is that its output can be compared systematically against a reference. This section reports the results of validating a 100,000-policy SynthDrive portfolio—generated with `random_state=42`, preset `core`, in seed mode—against the SBV seed dataset along four dimensions: marginal distributions, correlation structure, claim frequency relationships, and severity distribution. Section 5.5 summarizes all validation outcomes in a single table, distinguishing results that passed, those that are approximate, and those not tested in v0.1.

5.1 Marginal distributions

Figure 2 and Figure 3 show the marginal distributions of driver age and annual miles for the SynthDrive output and the SBV seed. Alignment is expected in seed mode, because SynthDrive uses the empirical quantile function of the SBV data directly as the marginal inverse CDF; the figures confirm that no material distortion is introduced by the copula sampling or the constraint-enforcement pass.

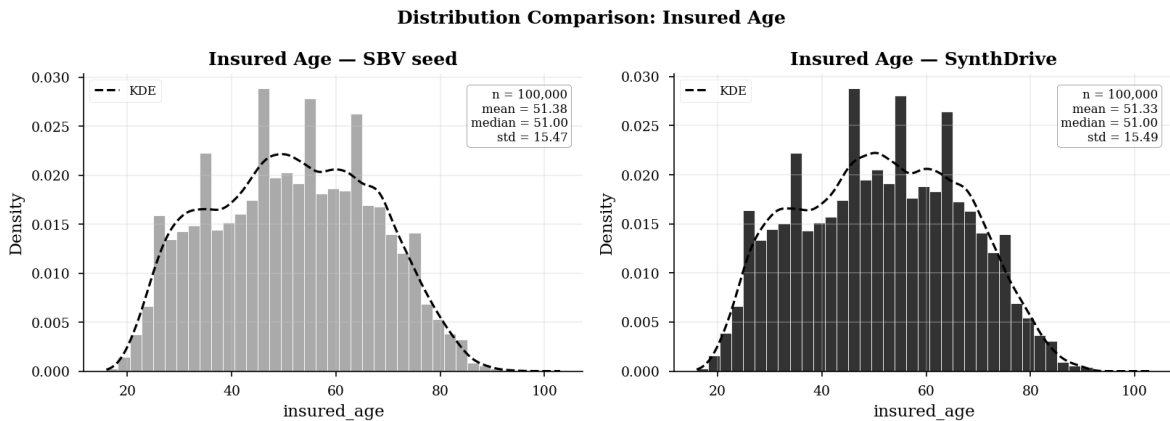


Figure 2: Marginal distribution of driver age: SBV seed (dashed gray) and SynthDrive output (solid dark gray), 100,000 policies each.

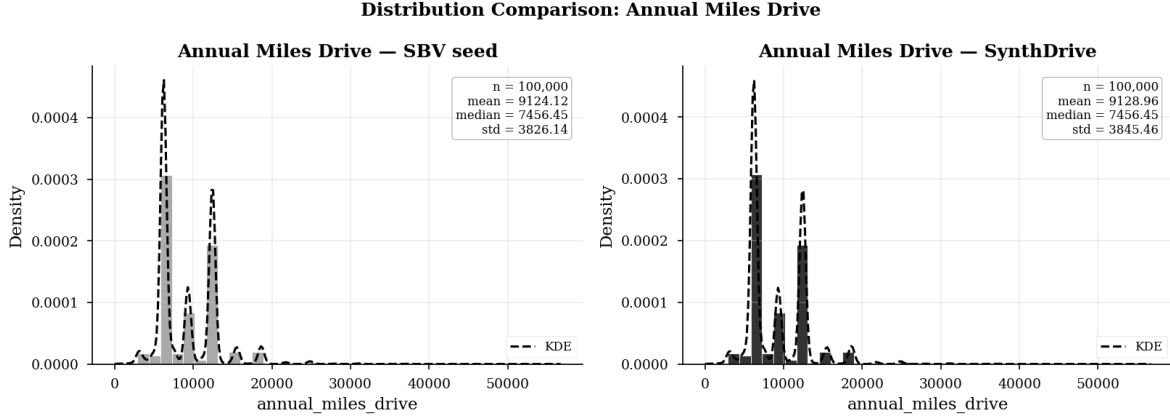


Figure 3: Marginal distribution of annual miles driven: SBV seed (dashed gray) and SynthDrive output (solid dark gray), 100,000 policies each.

5.2 Claim frequency

Figure 4 shows claim frequency by driver age band. The frequency model reproduces the broad actuarial shape: elevated frequency for young drivers (age < 25), declining through the main adult segment. SynthDrive continues to decline monotonically at older ages, whereas the SBV seed flattens into a plateau for drivers aged 56 and above; this divergence reflects the linear age parameterization in the risk score and is noted as an approximation in Table 5.

The portfolio mean claim frequency in the SynthDrive output (0.019 claims per policy) is modestly below the SBV seed mean (0.045), attributable to the conservative base annual rate $\lambda_0 = 0.0165$. This gap is disclosed as a known limitation in Section 6.

Figure 5 shows claim frequency by territory quartile. The SBV seed displays a clear upward trend from the lowest to the highest frequency quartile, whereas SynthDrive produces a nearly flat profile across quartiles. The territory effect in the risk score preserves rank ordering but does not reproduce the magnitude of the SBV gradient; this is noted as an approximation in Table 5.

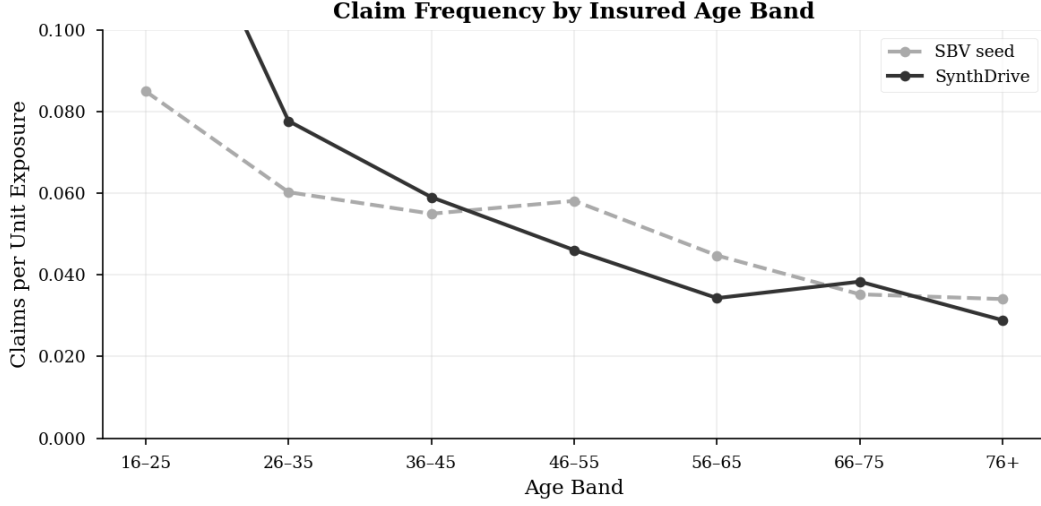


Figure 4: Claim frequency by driver age band: SBV seed and SynthDrive output, 100,000 policies each.

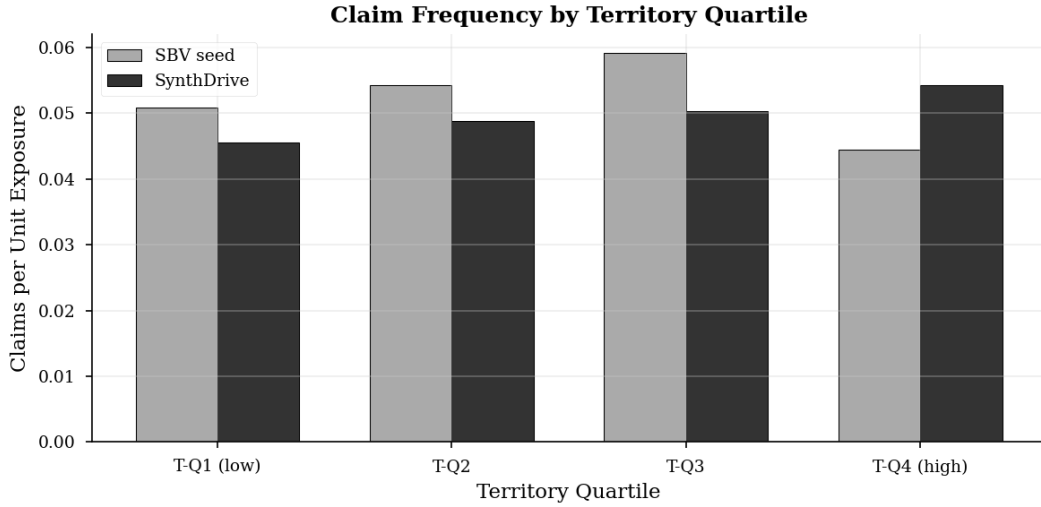


Figure 5: Claim frequency by territory quartile (territory codes sorted by frequency): SBV seed and SynthDrive output.

5.3 Claim severity

Figure 6 shows the log-severity distribution for policies with at least one claim, and Figure 7 shows the full claim amount distribution across all policies. The SynthDrive Gamma model, calibrated to a reference-driver expected severity of approximately \$3,134, reproduces the central mass of the log-severity distribution well. The upper tail is conservative: the single-component Gamma does not capture the heavy tail present in the SBV seed, where large bodily injury claims produce a visible right shoulder. This is a known limitation disclosed in Section 6.

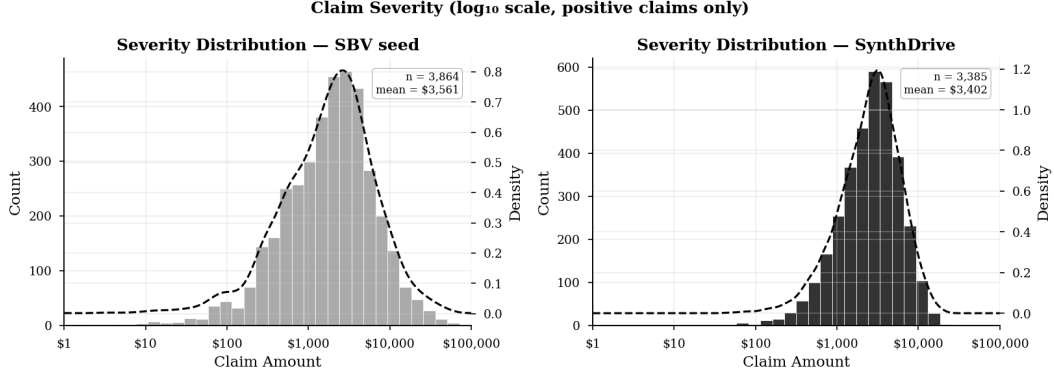


Figure 6: Log-claim-amount distribution for policies with at least one claim: SBV seed and SynthDrive output.

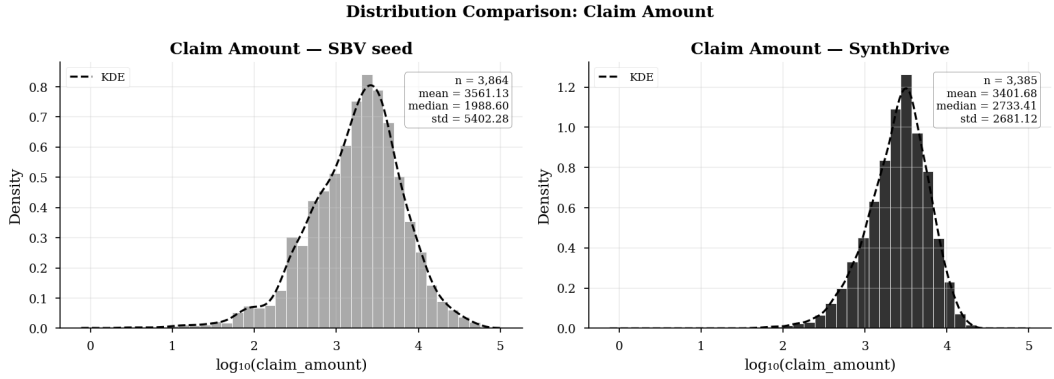


Figure 7: Claim amount distribution across all policies (zero for zero-claim policies): SBV seed and SynthDrive output.

5.4 GLM relativity comparison

Table 4 presents the central validation test: Poisson generalized linear models fitted separately to the SBV seed and the SynthDrive output, with coefficients compared across all main rating factors. The rightmost column displays each pair of estimates as an inline dumbbell on a fixed scale from -1.0 to $+3.0$, allowing direct visual comparison across rows. Five factors are well-calibrated to within 25%: vehicle age, credit score, log annual miles, young driver, and log hard braking. The annual percentage driven term has the largest absolute coefficient in both models and its estimates agree to within 2.3%. Terms marked $\dagger$ have near-zero coefficients in the SBV seed, making percentage comparisons uninformative; sign agreement is the appropriate criterion for these terms.

Table 4: Poisson GLM coefficient comparison: SBV seed vs. SynthDrive output (100,000 policies, `random.state=42`). Rows sorted by SBV estimate descending. Open circles: SBV seed. Filled circles: SynthDrive. Fixed scale -1.0 to $+3.0$. † SBV estimate below 0.05 in absolute value.

Variable	SBV	SynthDrive	Comparison
Annual pct. driven	+2.442	+2.499	●
Hard braking (log)	+0.462	+0.339	●○
Young driver (<25)	+0.426	+0.475	●
Log annual miles	+0.314	+0.273	●
Commercial use	+0.218	+0.428	○●
Senior driver (>65)†	+0.049	+0.002	●
Hard accel. (log)†	+0.035	+0.139	○●
Male†	+0.024	-0.030	●
Single†	-0.029	+0.057	○●
Vehicle age	-0.072	-0.073	●
Credit score (per 100 pts)	-0.307	-0.358	●
Farmer use	-0.383	-0.537	●○

5.5 Summary of validation outcomes

Table 5 summarizes the validation results, distinguishing tests that passed outright, those where the output is approximate but acceptable, and those not tested in v0.1.

Table 5: Validation summary for SynthDrive v0.1 (100,000-policy portfolio, seed mode, `random_state=42`).

Test	Outcome	Notes
Marginal distributions (continuous)	Pass	Seed-mode by design
Categorical proportions	Pass	Seed-mode by design
Territory distribution	Pass	Seed-mode by design
Compositional constraint (day-of-week)	Pass	Enforced by Dirichlet
Monotone hard-event sequences	Pass	Enforced by cummin
Claim consistency ($A_i = 0 \Leftrightarrow N_i = 0$)	Pass	Hard constraint
Exposure bounds	Pass	Hard constraint
GLM coefficient directions	Pass	All expected signs reproduced
Zero-claim fraction	Approximate	Within calibration target
Claim frequency by age band	Approximate	Sign and shape correct
Claim frequency by territory	Approximate	Rank ordering preserved
Log-severity distribution shape	Approximate	Gamma fit; heavy tail conservative
GLM coefficient magnitudes	Approximate	Well-calibrated for 6 of 13 terms (within $\pm 25\%$); near-zero denominator artifacts in 2 terms; magnitude gaps in remaining terms—see Table 4
Spearman correlation matrix	Approximate	Copula fit; off-diagonal moderate
Heavy-tail severity events	Approximate	Single Gamma; large-loss tail thin
Cross-country applicability	Not tested	SBV seed only; no non-Canadian data

All hard constraints pass by construction. The principal approximation gaps—overall claim frequency level and heavy-tail severity—are calibration choices made in the interest of transparency and are documented as limitations in Section 6.

6 Limitations

The following limitations apply to SynthDrive v0.1 and should be considered whenever the generated dataset is used in research.

Second-order synthetic provenance. SynthDrive is calibrated from the SBV synthetic seed, which was itself generated from a real but proprietary Canadian insurer dataset. SynthDrive output is therefore twice removed from real insurer data and should not be used to make claims about real insurance markets without independent validation against real data.

Single insurer, single jurisdiction. The SBV seed reflects the feature distributions and loss

statistics of a single Canadian insurer’s book at a point in time. The SBV authors do not disclose the policy year or the insurer’s portfolio characteristics in detail. SynthDrive makes no claim to reproduce the driving behavior or loss statistics of any other market. A Japan-calibrated variant is planned for v0.2, using published statistics from the General Insurance Association of Japan and related sources; that calibration will produce Japan-calibrated outcomes, not Japan-learned driving behavior, because no Japanese telematics data is used in training.

Gaussian copula limitations. The Gaussian copula models rank dependence but cannot reproduce tail dependence structures that may be present in real telematics data. In particular, the joint upper tail behavior of hard-braking and hard-acceleration counts—which may be relevant for fraud modeling—is modeled only approximately.

Linear risk score. The ZINB frequency model uses a parametric additive log-relative-risk score. Real insurer frequency models typically involve non-linear rate factors, interaction terms, and residual geographic structure that are not captured by the current linear score. The model is calibrated to reproduce the main actuarial relativities but is not designed to capture subtle non-linear patterns.

Single-component Gamma severity. The severity model is a single Gamma distribution with a modest risk adjustment on the scale parameter. Real claim severity distributions are typically heavier-tailed than a single Gamma, particularly in the presence of bodily injury claims. A mixture Gamma or Pareto mixture is planned for v0.2.

No privacy metrics. SynthDrive does not compute distance-to-closest-record or membership inference metrics. Because the seed is itself synthetic, the privacy threat model differs from the case of a generator trained on real data. Researchers should note that SynthDrive output is calibrated from a public dataset and carries no privacy guarantees beyond those of the SBV seed.

Classical methods only. SynthDrive is intentionally limited to classical statistical methods that run without GPU hardware. This prioritizes transparency and accessibility over the marginal fidelity gains that TabSyn and TabDDPM may offer on larger or more complex tabular datasets.

7 Conclusion

SynthDrive is a transparent, reproducible, and actuarially coherent synthetic telematics generator for motor insurance research. It addresses the data access problem in UBI research by providing an open-source tool that generates realistic telematics portfolios without requiring real insurer data. The design principles—classical statistical methods, enforced actuarial constraints, formal algebraic specification, built-in validation report—reflect a deliberate choice to prioritize inspectability and correctness over generative sophistication.

The central contribution is not a new modeling technique. Gaussian copulas, zero-inflated negative binomials, and Gamma severity models are well-established in the actuarial literature. The contribution is their assembly into a coherent, open, constraint-aware pipeline calibrated to a public

telematics seed, with a validation report that documents what the pipeline reproduces and what it does not.

The SBV dataset (So et al., 2021) was an important step in making telematics research accessible. SynthDrive is intended as a complement to it: a tool that researchers can run, inspect, modify, and calibrate without access to proprietary data, neural network training infrastructure, or a signed insurer data agreement. Extensions to additional generation methods and market calibrations are in development. The package code and formal specification are available at <https://github.com/nippotica/synthdrive>.

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References

- Bolderdijk, J.W., Knockaert, J., Steg, E.M., and Verhoef, E.T. (2011). Effects of Pay-As-You-Drive vehicle insurance on young drivers’ speed choice. *Accident Analysis and Prevention* 43(3), 1181–1186.
- Boucher, J.-P., Côté, S., and Guillen, M. (2020). Exposure as duration and distance in telematics motor insurance using generalized additive models. *Risks* 5(4), 54.
- de Jong, P. and Heller, G.Z. (2008). *Generalized Linear Models for Insurance Data*. Cambridge University Press.
- Frees, E.W. and Valdez, E.A. (1998). Understanding relationships using copulas. *North American Actuarial Journal* 2(1), 1–25.
- Higham, N.J. (2002). Computing the nearest correlation matrix—a problem from finance. *IMA Journal of Numerical Analysis* 22(3), 329–343.
- Klugman, S.A., Panjer, H.H., and Willmot, G.E. (2012). *Loss Models: From Data to Decisions*, 4th ed. John Wiley & Sons.
- Kotelnikov, A., Baranchuk, D., Rubachev, I., and Babenko, A. (2023). TabDDPM: Modelling tabular data with diffusion models. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, Honolulu, Hawaii.
- Kruskal, W.H. (1958). Ordinal measures of association. *Journal of the American Statistical Association* 53(284), 814–861.
- Lemaire, J., Park, S.C., and Wang, K.C.K. (2016). The use of annual mileage as a rating variable. *ASTIN Bulletin* 46(1), 39–69.
- Minami, M., Lennert-Cody, C.E., Gao, W., and Román-Verdesoto, M. (2007). Modeling shark bycatch: the zero-inflated negative binomial regression model with smoothing. *Fisheries Research* 84(2), 210–221.

- Nelsen, R.B. (2006). *An Introduction to Copulas*, 2nd ed. Springer.
- Paefgen, J., Staaake, T., and Fleisch, E. (2014). Multivariate exposure modeling of accident risk: Insights from pay-as-you-drive insurance data. *Transportation Research Part A: Policy and Practice* 61, 27–40.
- So, B., Boucher, J.-P., and Valdez, E.A. (2021). Synthetic dataset generation of driver telematics. *Risks* 9(4), 58. <https://doi.org/10.3390/risks9040058>
- Verbelen, R., Antonio, K., and Claeskens, G. (2018). Unravelling the predictive power of telematics data in car insurance pricing. *Journal of the Royal Statistical Society, Series C* 67(5), 1275–1304.
- Xu, L., Skoularidou, M., Cuesta-Infante, A., and Veeramachaneni, K. (2019). Modeling tabular data using conditional GAN. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vancouver.
- Zhang, H., Zhang, J., Lee, J., and Whang, S.E. (2024). Mixed-type tabular data synthesis with score-based diffusion in latent space. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, Vienna.
- Zhao, Z., Kunar, A., Birke, R., and Chen, L.Y. (2021). CTAB-GAN: Effective table data synthesizing. In *Proceedings of the 13th Asian Conference on Machine Learning (ACML)*.